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Research paper

Optimizing rice yield prediction: A policy-oriented approach to agricultural sustainability

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ABSTRACT

Accurate forecasting of rice yield is critical to ensuring global food security, supporting market stability, and enabling data-driven agricultural policy. This study evaluates a comprehensive suite of machine learning (ML) and statistical regression models to predict rice yield using meteorological and agronomic data from Kerala, India. Models assessed include Random Forest Regression (RFR), Gradient Boosting Regression (GBR), Support Vector Regression (SVR), Multiple Linear Regression (MLR), Least Absolute Shrinkage and Selection Operator (LASSO) Regression, Ridge Regression (RR), Elastic Net Regression (ELNR), and a hybrid ensemble combining LASSO, XGBoost, and RFR. GBR achieved the highest predictive accuracy (R^2 = 0.839), followed closely by SVR (R^2 = 0.837) and the hybrid model (R^2 = 0.827), with the hybrid model attaining the lowest Root Mean Square Error (RMSE). Beyond predictive performance, the study integrates causal inference to assess the impact of a policy intervention initiated in 2010. Regression results indicate that the policy is associated with a statistically significant yield increase of approximately 279 Kg/Ha ($p < 0.001$), while excessive rainfall is negatively associated with yield (–0.243 Kg/Ha/mm; $p < 0.001$). The integration of predictive analytics and policy-aware modeling presents a robust framework for forecasting yield and evaluating the real-world effectiveness of agricultural interventions. These insights offer substantial value for policymakers and stakeholders aiming to optimize food production systems.

1. Introduction

Agricultural productivity is a key determinant of economic growth and food security. It refers to the output generated per unit of land, labor, and capital in agriculture. Over time, productivity has been enhanced through the adoption of improved farming techniques, efficient resource management, and modern technologies such as precision agriculture and data analytics. These innovations improve yields while reducing environmental impacts and increasing resilience to climate change. Through better fertilizer use, water management, and pest control, farmers can maximize production sustainably, ensuring greater food availability and improved livelihoods for agricultural communities. Higher productivity also contributes to poverty reduction, supports rural development, and remains a cornerstone of sustainable agriculture.

The agricultural sector has benefited significantly from recent advances in sensor technology, data science, and Machine Learning (ML) [1]. These innovations aim to address pressing environmental and demographic challenges. Reports indicate an urgent need to expand global agricultural productivity to meet the food demands of a growing population in a warming climate. The dynamic and unpredictable environment marked by frequent deviations from established weather patterns presents a considerable challenge. Advanced technologies can enhance the efficient use of key resources such as land, water, and air, thereby improving yields and helping combat food insecurity.

Computer models can simulate the impacts of climatic conditions on crop growth, accounting for extreme events such as heatwaves, heavy rainfall, and waterlogging. Such tools help bridge the gap between traditional farming practices and modern agricultural technology. Experimental data allow researchers to identify environmental zones affected by rainfall variation a critical factor in successful crop production [4]. By simplifying complex real-world conditions, researchers can more effectively evaluate the impacts of rainfall variability on agriculture.

Crop yield estimation is a critical component of modern agriculture. It combines historical data with current environmental parameters to forecast future harvests [5]. This study focuses on the urgent need for

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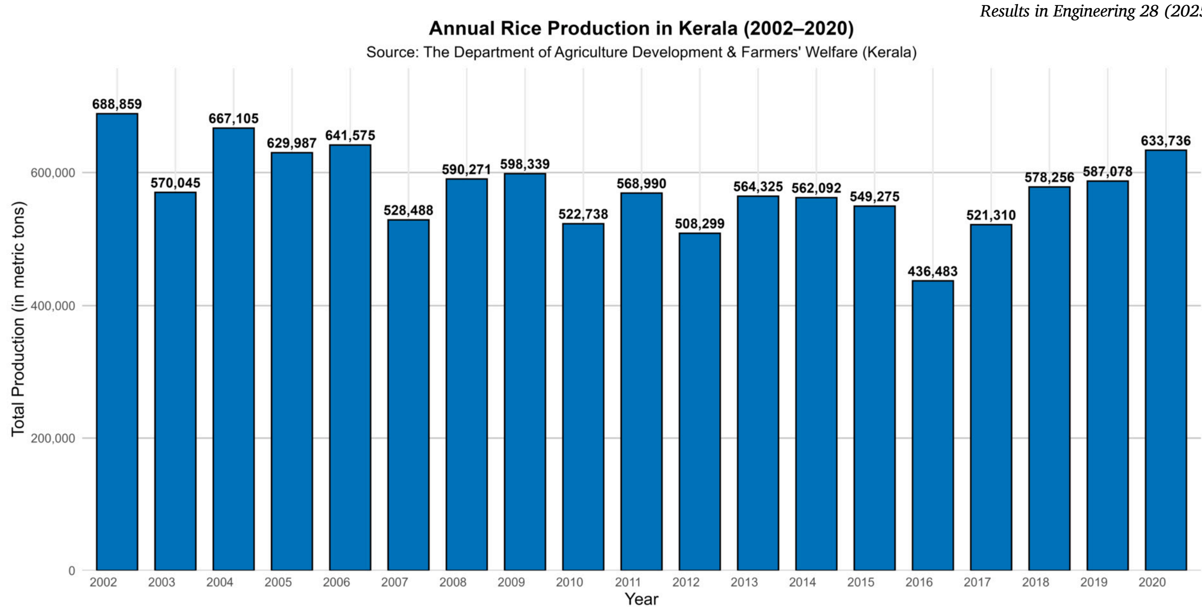


Fig. 1. The rice productivity in Kerala from 2002 to 2020.

accurate and actionable yield predictions in Kerala, a state facing declining rice productivity despite its central role in food security. By applying advanced statistical methods and ML models, policymakers and agricultural professionals can make informed decisions about resource allocation, improve productivity, and strengthen food security [24,9]. Accurate crop yield forecasting is thus essential for the stability and growth of the agricultural sector.

Rice is the world's second most important cereal crop following only corn [25]. Rice serves as the primary nutrition for more than 50% of the global population and has played a substantial role in ensuring food security worldwide [28]. In addition, rice is grown on a worldwide scale across 156 million hectares, with over 85% of this cultivation taking place in Asia [2].

Kerala's agricultural landscape presents unique challenges for yield prediction due to variable climatic conditions, fragmented landholdings, and diverse cropping patterns. Traditional prediction methods often fail to account for non-linear dependencies and multicollinearity among variables such as rainfall, temperature, soil quality, and irrigation. Advanced regression techniques and hybrid machine learning models offer solutions by capturing complex relationships within the data.

In India, about 60% of land is used for agriculture, and rice cultivation is widespread across nearly every state. In recent years, modern technologies have been increasingly adopted to improve agricultural efficiency, offering the potential for higher farmer incomes nationwide.

Fig. 1 highlights the importance of accurate rice yield forecasts for Kerala. Such predictions are essential for food security, economic planning, climate adaptation, resource management, and technology integration. By analyzing historical yield data alongside weather patterns, predictive models can help design strategies to adapt to climate change while maintaining sustainable farming practices. These forecasts also enable farmers to optimize input use, reduce waste, and minimize environmental impacts. Advances in statistical and ML approaches provide more reliable predictions, supporting policy development to address agricultural challenges.

Rice is the principal food crop cultivated in Kerala, covering 7.46% of the total cropped area. However, the land dedicated to rice has sharply declined since the 1980s—from 8.82 lakh hectares in 1974–75 to just 1.96 lakh hectares in 2015–16. Production has also dropped from 13.76 lakh MT in 1972–73 (peak output) to 5.49 lakh MT in 2015–16. For comparison, China, the world's largest rice producer, achieves yields of 6,744 kg/ha—more than three times Kerala's output. Egypt records the world's highest rice productivity at 9,088 kg/ha, nearly four times that of Kerala. Restoring rice production in the state requires proactive measures with strong support from both the Department of Agriculture and local authorities.

Fig. 1 shows Kerala's rice yields between 2002 and 2020. Over this 19-year period, yields fluctuated but displayed a general upward trend. Starting at 30,829 MT in 2002, yields dipped slightly in 2003 before recovering in subsequent years. Notable peaks occurred in 2013 and 2017, both surpassing 36,000 MT. The highest yield was recorded in 2018, at 37,385 MT. Despite periodic declines, the overall trajectory indicates improvements in farming practices or favorable climatic conditions, underscoring the potential for continued growth and stability in rice production.

Kerala is located in southwestern India, bordered by the Arabian Sea to the west and the Western Ghats to the east, spanning an area of 38,863 square kilometers. The state comprises 14 districts with diverse topographical and climatic conditions, strongly influencing agricultural practices and crop yields. Agriculture plays a central role in Kerala's economy, employing a significant portion of the population. Paddy fields dominate the lowlands, while the highlands are known for spices and plantation crops. These geographical and climatic variations necessitate region-specific farming practices to improve yield and sustainability. Fig. 2 shows Kerala's geographical location [16]. The state lies between 8°17'30" and 12°47'40" north latitude and 74°27'47" to 77°37'12" east longitude. Kerala experiences a tropical climate, with average temperatures ranging from 28 °C to 32 °C and annual rainfall of around 3,000 mm.

1.1. Research aim and objectives

The primary objective of this study is to identify the most relevant indicators of field variability for rice yield prediction. The specific objectives are:

1. To identify critical indicators of field variability that significantly impact rice yield.
2. To examine the influence of historical rainfall patterns and other environmental factors on yield.
3. To develop and evaluate hybrid models that integrate statistical and ML techniques for yield prediction.
4. To demonstrate the advantages of hybrid approaches over single models in achieving high prediction accuracy and generalization.

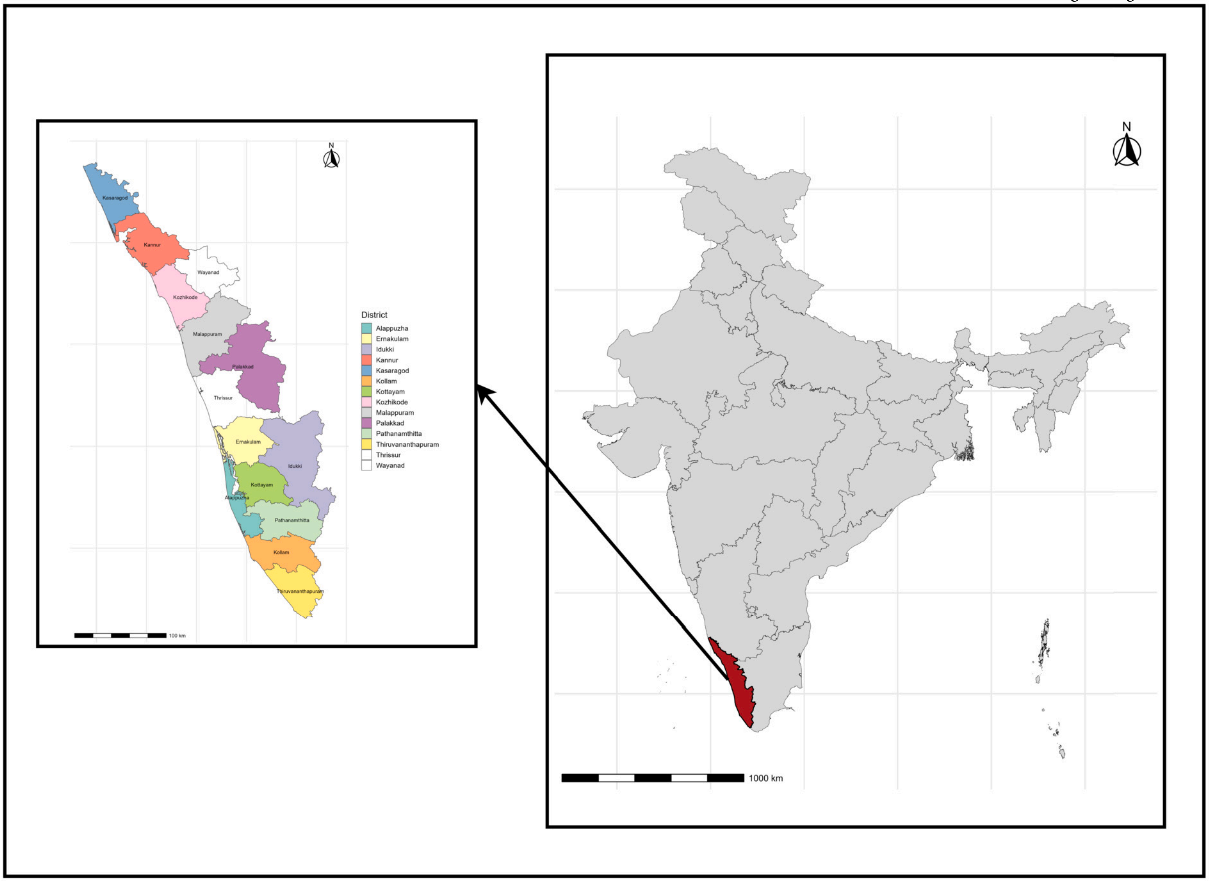


Fig. 2. Spatial overview of the study area, showing its geographical boundaries and land use patterns. These visuals provide contextual grounding for subsequent yield and policy analyzes.

This study uses 2010 as a structural breakpoint to assess changes in rice yield trends. This choice is based on two considerations:

- It represents the mid-point of the dataset (2002–2020), allowing for balanced pre- and post-comparisons, and
- Around 2010, Kerala's agricultural sector underwent gradual institutional and technological shifts, including expanded mechanization, decentralized procurement schemes, and increased investment in irrigation.

It is important to note that 2010 does not correspond to a single policy change. Instead, it serves as a proxy for a broader transition in rice farming practices and support mechanisms. The "post-2010" period thus reflects the cumulative effects of multiple institutional, technological, and environmental factors, rather than the outcome of a single intervention.

The study will be structured as follows: Section 2 provides a detailed literature review highlighting the significance of our research; Section 3 elaborates on the methodologies employed, divided into four primary components 3.1, data sources and research area; 3.2, machine learning models; 3.3, model evaluation; and 3.4, exploratory data analysis and Section 4 presents the results and discussion based on the analysis. Section 5: Conclusion and future scope of the study. Lastly, the references that provide the basis of our study are given.

2. Literature survey

The application of machine learning (ML) techniques in crop yield prediction has gained significant traction in recent years, particularly in addressing challenges posed by climate variability, food security, and sustainable agriculture. This review synthesizes key studies that integrate ML models with climatic, remote sensing, and agronomic data to forecast yields for major crops such as rice, wheat, and others. These studies span regions including South Asia (Pakistan, India, Sri Lanka) and East Asia (China), highlighting the critical role of environmental factors like temperature, rainfall, and humidity in influencing crop yields. Common ML algorithms employed include Random Forest (RF), Gradient Boosting Regression (GBR), Artificial Neural Networks (ANN), Support Vector Machines (SVM), and regression-based models. These approaches consistently outperform traditional statistical methods by effectively capturing non-linear relationships among variables. However, challenges such as limited data availability, model scalability, and the need for region-specific adaptations remain. The following subsections summarize key findings from the literature, followed by a comparative table for visualization and an identified research gap.

2.1. Key studies on ML-based crop yield prediction

Ashfaq et al. [4] conducted a study in the Multan district of Pakistan, utilizing ML and climate-NDVI data fusion to predict wheat yields. The RF model achieved an impressive accuracy of 97%, outperforming other ML techniques. The integration of climate and remote sensing data significantly enhanced prediction accuracy compared to single-source approaches, with critical periods identified from late February to March, corresponding to the grain-filling phase. A negative correlation between rainfall and yield was noted, attributed to issues like pre-harvest sprouting. Data were sourced from Punjab's crop reporting services and the Pakistan Meteorological Department. The study highlighted RF's ability to capture non-linear interactions and recommended further exploration at larger regional scales to address changing climatic conditions.

Hoque et al. [14] developed a framework for yield prediction of major Indian crops (rice, wheat, potatoes, soybeans, sweet potatoes, and sorghum) using ML models: Yield Multiple Linear Regression (YMLR), Yield K-Nearest Neighbors (YK-NN), and Yield Gradient Boosting Regression (YGBR). YGBR demonstrated the highest predictive accuracy. Strong positive correlations were observed between meteorological parameters (temperature and rainfall) and crop yields. Optimal conditions for potato yield included an average temperature of 26.2 °C, 1120 mm rainfall, and timely pesticide application. Data integration from FAO and World Bank sources via extract-transform-load procedures addressed challenges in reliable data acquisition, underscoring the need for tailored environmental management strategies to optimize productivity.

Panigrahi, et al. [17] compared ML algorithms (RF, Support Vector Regression (SVR), K-Nearest Neighbors (KNN), XGBoost (XGB), GBR, and Decision Tree (DT)) for rice and sugarcane yield prediction in Gujarat, India. RF and GBR performed best for rice, while GBR combined with XGBoost excelled for sugarcane. However, Mean Absolute Percentage Error (MAPE) exceeded 8% across models, indicating room for improvement by incorporating temperature and humidity features. The authors emphasized continuous model validation to enhance agricultural planning and decision-making.

Satpathi et al. [20] conducted a comparative analysis of rice yield forecasting in Chhattisgarh, India, using statistical and ML techniques. ML models, particularly ANN, outperformed traditional statistical methods in accuracy and predictive power. Weighted weather indices proved more effective than simple ones, underscoring the influence of weather on yields. The study advocated for accessible data and region-specific refinements to support food security, resource management, and policy-making amid climate variability.

Setiya et al. [21] evaluated four regression techniques for rice yield prediction in Udham Singh Nagar, Uttarakhand, India. ANN achieved the best performance in terms of R^2, Root Mean Square Error (RMSE), and Mean Absolute Error (MAE). Key weather parameters included maximum and minimum temperature, rainfall, relative humidity, and solar radiation. Tests for residual normality, homoscedasticity, and multicollinearity validated the results, suggesting ANN's potential scalability for broader crop-weather forecasting.

Wickramasinghe et al. [27] applied multiplicative modeling techniques (ANN, SVM Regression (SVMR), Multiple Linear Regression (MLR), Gaussian Process Regression (GPR), Polynomial Regression (PR), and Ridge Regression (RR)) for rice yield forecasting in Kurunegala and Puttalam districts, Sri Lanka. GPR models were the most accurate, incorporating variables such as rainfall, temperatures, evaporation, wind speed, and sunshine hours. Yields varied by season (Yala and Maha) and district, with performance metrics (MSE, R, MAPE, etc.) confirming model efficacy for agricultural planning and food security.

Guo et al. [13] examined the impact of climate change on rice yields in South China, using data from 75 experimental stations (1981–2010). Employing SVM, RF, and MLR, they found phenological variables to be more critical than climatic ones, with preseason climatic factors enhancing accuracy. ML methods outperformed traditional approaches, emphasizing the need for integrated data sources to improve predictions under climate change.

Shastry et al. [23] implemented regression models (multiple linear, quadratic, pure-quadratic, interactions, and polynomial) for wheat, maize, and cotton yield prediction in India. Simple Linear Regression (SLR) performed best for maize and cotton based on RMSE, R^2, and Mean Percentage Prediction Error (MPPE). The study highlighted the importance of attribute selection and suggested exploring SVM, neural networks, and fuzzy logic to further improve accuracy.

Yumlembam et al. [31] used linear regression for rice yield forecasting in Imphal West, Manipur, India. Positive correlations were found between yields and climatic variables (rainfall, temperature, humidity, sunshine), with sunshine hours being most influential. Declining rainfall trends posed threats to rice cultivation, while fertilizer use was linked to increased greenhouse gas emissions. The study recommended adaptation and mitigation strategies, including sustainable agricultural practices to reduce emissions.

Gandhi et al. [11] proposed an ML-based model for Indian crops (rice, wheat, potatoes, soybeans, sweet potatoes, sorghum), mirroring findings from Hoque et al. [14]. YGBR achieved the highest accuracy, with optimal potato conditions at 26.2 °C and 1120 mm rainfall. Data integration from FAO and World Bank sources improved predictions despite challenges in data reliability, highlighting the need for environmental management strategies.

Aselisewine [3] concluded that combining diverse datasets enhances predictive accuracy, but region-specific factors and continuous validation are critical. The study recommended scalable models incorporating high-resolution data (e.g., real-time weather) and novel features to support sustainable agricultural practices.

2.2. Comparative visualization of existing studies

While recent studies (Ashfaq et al., 2024; Setiya et al., 2024) have explored machine learning applications in agricultural forecasting, our study offers distinct contributions. First, we introduce a hybrid ensemble framework that integrates multiple regression models to enhance predictive robustness. Second, we explicitly model district-level heterogeneity, enabling localized insights often overlooked in regional analyzes. Third, we operationalize ML-causal integration by linking policy intervention variables with yield outcomes in a statistically validated framework. These elements collectively advance the methodological frontier and support scalable, policy-relevant planning for sustainable agriculture.

To provide a clear comparison, Table 1 summarizes the reviewed studies, detailing authors, crops, regions, methods, key findings, and limitations. This facilitates the identification of trends and gaps in ML applications for yield prediction.

2.3. Identified research gap

Despite significant progress in applying ML to crop yield prediction, several gaps remain unaddressed. Current studies demonstrate ML's ability to capture non-linear climate–crop interactions; however, limitations exist in crop coverage, data quality, validation, and methodological advancements. These gaps hinder the scalability, accuracy, and generalizability of existing models.

- Most research is confined to specific regions and a narrow set of crops (mainly rice and wheat), with limited exploration of multi-crop models at broader national or global scales.
- Inconsistent data sources, lack of real-time and high-resolution inputs, and insufficient integration of satellite or sensor-based data contribute to higher prediction errors, with MAPE values often exceeding acceptable thresholds (such as, > 8% in [17]).
- Few studies incorporate climate change projections, IoT-sensor data, or other advanced features, restricting the ability of models to address future climate uncertainty and evolving agricultural practices.
- Validation across diverse agro-climatic zones remains inconsistent, while hybrid approaches combining ML with deep learning or ensemble techniques are underexplored, limiting scalability, generalizability, and robustness of predictions.

3. Methodology

This study employs a comprehensive machine learning framework to predict rice yield using a dataset comprising agricultural and environmental variables [7]. The methodology encompasses data preprocessing, model development, hyperparameter tuning, and performance evaluation. Multiple regression models, including Random Forest (RF), Gradient Boosting Machine (GBM), Support Vector Regression (SVR), Multiple Linear Regression (MLR), LASSO regression, Ridge Regression

Table 1
Comparative Analysis of ML-Based Crop Yield Prediction Studies.

Author(s) & Year	Crop(s)	Region	Key Methods	Key Findings	Limitations/Suggestions
Ashfaq et al. (2024)	Wheat	Multan, Pakistan	RF, ML; climate-NDVI fusion	97% accuracy with RF; negative rainfall-yield correlation; critical period Feb–Mar	Larger-scale studies needed
Hoque et al. (2024)	Rice, wheat, potatoes, etc.	India	YMLR, YK-NN, YGBR	YGBR best; optimal potato conditions (26.2 °C, 1120 mm rain)	Data acquisition challenges; region-specific strategies
Virani et al. (2023)	Rice, sugarcane	Gujarat, India	RF, SVR, KNN, XGB, GBR, DT	RF/GBR best for rice; GBR+XGB for sugarcane; MAPE >8%	Add temp/humidity; continuous validation
Satpathi et al. (2023)	Rice	Chhattisgarh, India	Statistical vs. ML (ANN)	ANN superior; weighted weather indices effective	Improve data accessibility; region-specific refinements
Setiya et al. (2024)	Rice	Udham Singh Nagar, India	Regression (ANN)	ANN best (R^2, RMSE, MAE); key weather params	Validate assumptions; scale to larger areas
Wickramasinghe et al. (2021)	Rice	Kurunegala, Puttalam, Sri Lanka	ANN, SVMR, MLR, GPR, PR, RR	GPR most accurate; seasonal/district variations	More climate data for global strategies
Guo et al. (2021)	Rice	South China	SVM, RF, MLR	Phenological vars key; preseason factors improve accuracy	Integrate diverse data for climate resilience
Shastry et al. (2017)	Wheat, maize, cotton	India	Various regressions (SLR)	SLR best for maize/cotton; attribute selection crucial	Explore SVM/ANN; analyze correlations
Yumlembam et al. (2017)	Rice	Imphal West, India	Linear regression	Sunshine most influential; declining rainfall threat	Adaptation policies; reduce emissions
Gandhi et al. (2016)	Rice, wheat, potatoes, etc.	India	YMLR, YK-NN, YGBR	YGBR best; similar to Hoque et al.	Data challenges; environmental management
Aselisewine (2024)	General crops	Global/India	ML integration review	Dataset fusion boosts accuracy; scalable models needed	High-res data, real-time features

(RR), Elastic Net Regression (ELNR), and a hybrid ensemble model, were implemented to ensure robust predictions. The following subsections detail the data preparation, model specifications, and evaluation metrics (Fig. 3).

3.1. Exploratory data analysis

Exploratory Data Analysis (EDA) was conducted to understand the structure, distribution, and relationships within the rice yield dataset (`rice_yield_data.csv`). The dataset comprises 266 observations across six variables: district (categorical), year (numeric), area (numeric, in hectares), production (numeric, in tonnes), yield (numeric, in kg/ha), and actual rainfall (numeric, in mm). The following subsections provide a detailed interpretation of the summary statistics and distributional properties, facilitating the understanding of the data's characteristics and informing subsequent modeling decisions.

3.2. Data overview

The dataset used in this study is structured as cross-sectional time-series (panel data), comprising multiple districts observed over several years. To analyze the relationship between crop yield and multiple explanatory variables such as cultivated area, rainfall, and policy interventions a pooled panel regression approach was adopted using MLR. This method treats all observations as part of a single dataset, assuming consistent effects across districts and time.

In this study, rainfall was selected as the primary explanatory variable for rice yield forecasting. This decision was based on the availability of consistent, district-level rainfall data across the full study period (2002–2020), whereas other agroclimatic indicators such as temperature, humidity, and soil fertility were either unavailable or incomplete at comparable spatial and temporal resolutions. Furthermore, rainfall is agronomically critical for rice in Kerala, where cultivation is highly sensitive to monsoon variability, waterlogging, and irrigation dependence. While this narrow focus provides a parsimonious and robust modeling framework, we acknowledge that excluding additional climatic and edaphic variables is a limitation.

Table 2 discusses dataset includes data from 14 districts in Kerala, India, with each district represented by 19 observations, indicating a balanced representation across districts. The data spans from 2002 to 2020, covering 18 years, with a mean and median year of 2011, suggesting a symmetric temporal distribution (skewness = 0.00, kurtosis = -1.22). The absence of skewness in the year variable confirms a uniform spread of observations over the study period, which is critical for temporal consistency in predictive modeling.

Outliers were identified in the dataset using the interquartile range (IQR) method. Specifically, 20 outliers were detected in the Area variable, 21 in Production, and 11 in Yield. No outliers were found in Actual Rainfall, indicating a relatively consistent distribution of rainfall across districts and years.

3.3. Distribution of key variables

The Table 3 shows area variable, representing the cultivated land in hectares, exhibits significant variability, ranging from 616 to 115,910 hectares (mean = 16,200.74, SD = 23,266.27). The distribution is right-skewed (skewness = 2.58) with a high kurtosis (6.42), indicating a leptokurtic distribution with heavy tails and a concentration of smaller values, alongside a few extremely large values. This suggests that most districts cultivate relatively smaller areas, with a few outliers having substantially larger cultivated areas, potentially influencing yield predictions.

Production, measured in tonnes, similarly shows a wide range (1,467 to 270,103 tonnes, mean = 41,155.08, SD = 59,911.27) and is right-skewed (skewness = 2.40, kurtosis = 5.24). The high variability and skewness indicate that production varies significantly across districts, likely driven by differences in cultivated area and agricultural practices. The median production (15,510.5 tonnes) is notably lower than the mean, reinforcing the presence of outliers with high production values.

The target variable, yield (kg/ha), ranges from 1,256 to 3,685 kg/ha (mean = 2,421.05, SD = 460.48). The distribution is approximately symmetric (skewness = -0.16) with a moderate kurtosis (0.25), suggesting a near-normal distribution. This relatively stable distribution, with a median (2,385 kg/ha) close to the mean, indicates consistent yield

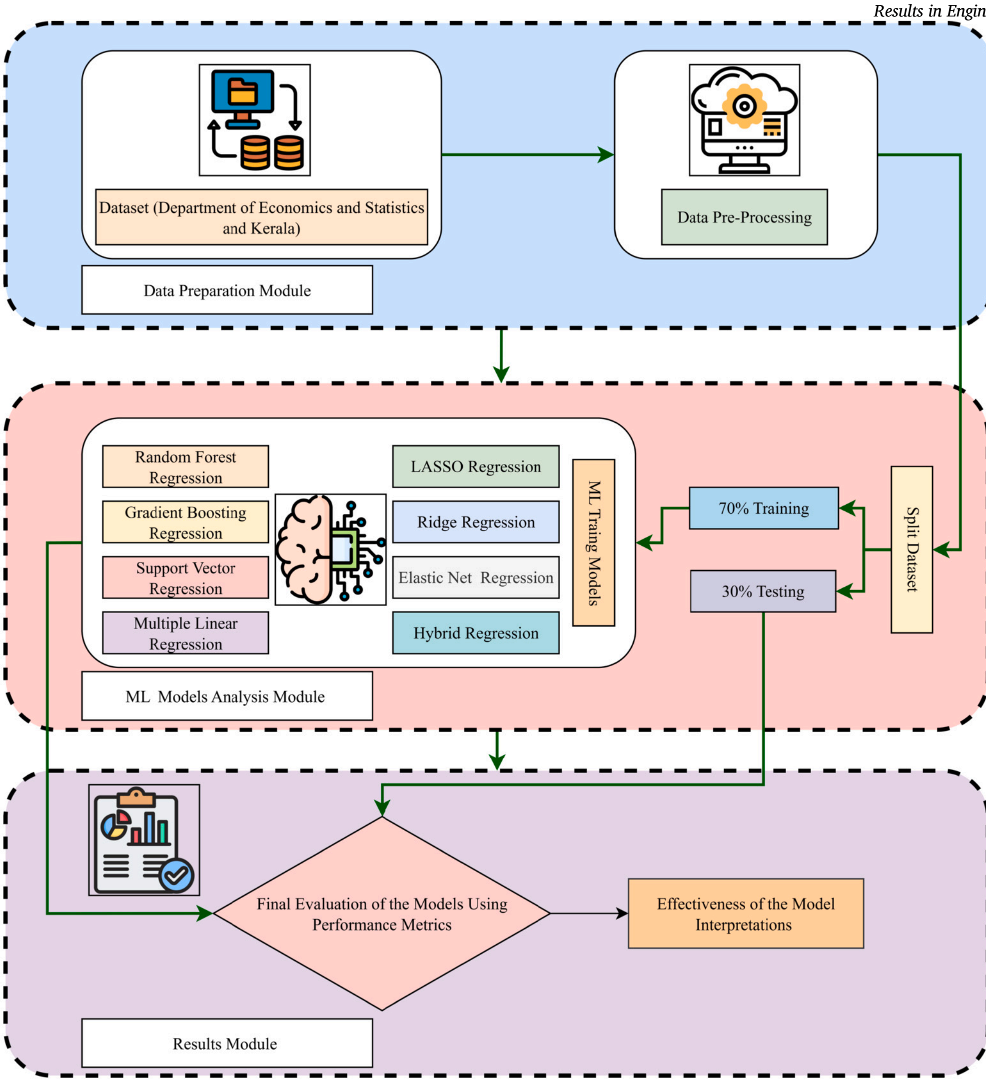


Fig. 3. Machine Learning Workflow Optimization for Enhanced Agricultural Insights.

Table 2
Description of agricultural dataset.

Feature	Explanation	Range	Measurement
District	This column indicates the name of the district where the data was collected. The dataset includes districts like Thiruvananthapuram, Kollam, Pathanamthitta, Alappuzha, Kottayam, Idukki, Ernakulam, Thrissur, Palakkad, Malappuram, Kozhikode, Wayanad, Kannur, and Kasaragod.	1 to 14	Na
Year	This represents the year of the data.	2002 to 2020	days
Area	This variable represents the cultivated area in hectares.	616 to 115910	Hectares (Ha)
Production	This column indicates the total crop production measured in metric tons.	1467 to 270103	Tonnes (Kg)
Yield	Yield is a measure of productivity, calculated as the amount of crop produced per unit area. It is usually expressed in kilograms per hectare (kg/ha).	1256 to 3685	Kilograms per Hectare (Kg/Ha)
Actual Rainfall	This variable provides the actual amount of rainfall received in millimeters(mm) during the year.	1505 to 3361	Millimeters (mm)

Table 3
The statistical descriptive statistics of the four variables.

Statistical description	Area (Ha)	Production (Tonnes)	Yield (Kg/Ha)	Actual Rainfall (mm)
Mean	16200.74	41155.08	2421.05	2517.57
Standard Deviation	23266.27	59911.27	460.48	521.09
Median	7207	15510.5	2385	2523.5
Skewness	2.58	2.4	-0.16	-0.16
Kurtosis	6.42	5.24	0.25	-0.86
Median Absolute Deviation	7099.43	16046.18	378.06	626.4

patterns across observations, making it suitable for regression modeling. The standard error (SE = 28.23) suggests moderate precision in the mean estimate.

Actual rainfall, measured in millimeters, ranges from 1,505 to 3,361 mm (mean = 2,517.57, SD = 521.09). The distribution is slightly left-skewed (skewness = -0.16) with a platykurtic shape (kurtosis = -0.86), indicating a relatively uniform spread with fewer extreme values compared to area and production. The median rainfall (2,523.5 mm) aligns closely with the mean, and the standard error (SE = 31.95) suggests reliable estimation. Rainfall is a critical environmental factor influencing rice yield, and its moderate variability suggests consistent climatic conditions across the study period.

3.4. Data preprocessing

Data preprocessing was performed using the `tidyverse` package in R. The district variable was converted to a categorical factor, while area, production, yield, and actual rainfall were treated as numeric variables. The first column, assumed to be an identifier, was excluded from analysis. The dataset was split into training (70%) and testing (30%) sets using random sampling with a fixed seed (`set.seed(222)`) to ensure reproducibility. The training set was used for model development, while the testing set was reserved for performance evaluation.

3.5. Model development

Seven machine learning models and a hybrid ensemble model were developed to predict rice yield. Each model was trained using the `caret` package in R, with 5-fold cross-validation to optimize hyperparameters and mitigate overfitting. The models are described below, along with their respective hyperparameter tuning strategies.

3.5.1. Random Forest (RF)

Random Forest regression is an ensemble learning method popular for regression problems, whereby during training, an ensemble of trees is created. The final prediction is obtained by taking an average over all the trees. It boosts the predictive performance and reduces overfitting due to collective wisdom, since increasing the diversity of trees decreases variance and increases the model's robustness [18].

The RF model, implemented via the `randomForest` package, constructs an ensemble of decision trees to predict yield. The number of trees was set to 500, and the number of features randomly sampled at each split (`mtry`) was tuned over the values $\{2, \lfloor p/3 \rfloor, p-1\}$, where p is the number of predictors. The RF model minimizes the Mean Squared Error (MSE), defined as:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2, \tag{1}$$

Where y_i is the observed yield, $\hat{y}_i$ is the predicted yield, and n is the number of observations.

3.5.2. Gradient Boosting Machine (GBM)

Gradient boosting is a widely used method in Machine Learning (ML). Applications such as regression and classification can utilize this technique. The technique generates a prediction model from a collection of weak prediction models, with decision trees being the most commonly used due to their strong predictive capabilities. In some cases, a decision tree is used to represent the weak learner, and the strategy developed such a system using gradient-boosted trees [19]. This method frequently delivers results that outperform RF.

The GBM model, implemented using the `gbm` package, builds an ensemble of trees sequentially, optimizing a loss function. Hyperparameters tuned include the number of trees (`n.trees = {500, 1000}`), interaction depth (`interaction.depth = {1, 3}`), shrinkage rate (`shrinkage = {0.01, 0.1}`), and minimum observations per node (`n.minobsinnode = 10`). The loss function minimized is the squared error loss, as defined in Equation (1).

3.5.3. Support Vector Regression (SVR)

Support Vector Regression (SVR) is a form of Support Vector Machine (SVM) that applies its concepts to regression tasks [30]. SVR seeks to identify a function that differs from the actual observed values by an amount not exceeding a defined margin (epsilon). It employs a kernel function to convert the input data into a higher-dimensional space, enabling linear regression to be executed. The goal of SVR is to reduce the model's complexity by achieving a balance between the margin of tolerance and model simplicity. This renders SVR a robust method for modeling intricate, non-linear relationships within the data.

SVR, implemented via the `kernlab` package with a Radial Basis Function (RBF) kernel, was customized to include the epsilon parameter. The hyperparameters tuned were the cost parameter (`C = {0.1, 1, 10}`), kernel parameter (`sigma = {0.01, 0.1, 1}`), and epsilon (`epsilon = {0.01, 0.1, 0.5}`). The SVR model minimizes the ϵ-insensitive loss function:

$$L_\epsilon(y_i, \hat{y}_i) = \begin{cases} 0 & \text{if } |y_i - \hat{y}_i| \le \epsilon, \\ |y_i - \hat{y}_i| - \epsilon & \text{otherwise,} \end{cases} \tag{2}$$

subject to a regularization term controlled by `C`.

The optimal configuration C = 10, sigma = 0.01, and epsilon = 0.1 achieved the best predictive performance with RMSE = 165.99, MAE = 120.62, and R^2 = 0.878. This indicates that the model effectively captured non-linear relationships between predictors and yield while maintaining generalization across test data.

3.5.4. Multiple Linear Regression (MLR)

Multiple Linear Regression (MLR) is a broader form of simple linear regression that analyzes the connection between two or more independent variables and one dependent variable. Its purpose is to forecast the value of the dependent variable using the values of the independent variables. The strength of MLR is its capacity to process multiple predictors at once, rendering it a robust tool for predictive modeling.

The MLR model assumes a linear relationship between predictors and yield, expressed as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon, \tag{3}$$

Where β_0 is the intercept, β_j are the coefficients, x_j are the predictors, and ϵ is the error term. The model was trained using ordinary least squares to minimize the residual sum of squares.

3.5.5. LASSO, ridge, and elastic net regression

LASSO

A regression analysis technique called Least Absolute Shrinkage and Selection Operator (LASSO) achieves regularization and variable selection to improve the model's interpretability and prediction accuracy [29]. It modifies the method of fitting the model and chooses only a portion of the variables. It does this by making certain coefficients 0 and deleting them from the model while it is being built.

Ridge

Ridge regression, also referred to as Tikhonov regularization, is a method utilized to tackle multicollinearity in linear regression models. Multicollinearity arises when predictor variables are closely correlated, resulting in unreliable estimates of regression coefficients [12]. Ridge regression incorporates a penalty term into the least squares objective function, thereby reducing the coefficients towards zero, though not completely to zero. This regularization enhances the model's ability to generalize by avoiding overfitting. The penalty term is regulated by a tuning parameter, lambda (λ), which influences the intensity of the regularization. The higher the λ, the more significant the shrinkage.

Elastic Net

In 2005, Zou and Hastie introduced the elasticity net regression. The described method is a hybrid regression methodology that includes penalized least squares regression, regularization, and variable selection. ELNR combines two useful shrinkage regression methods: ridge regression (L2 penalty) for dealing with high-multicollinearity difficulties, and LASSO regression (L1 penalty) for selecting the appropriate regression coefficients. ELNR reduces the coefficient regression to zero or equal to zero for less relevant predictor variables. This is accomplished by multiplying the sum of the absolute values of the coefficient variables (L1 norm) by the tuning parameter λ_1. To address significant correlation between predictor variables, the tuning parameter λ_2 is multiplied by the sum of squared coefficient variables (the L2 norm). The ELNR concept enhances the fitting model's interpretability by minimizing the number of irrelevant variables, resulting in more accurate predictions. ELNR reduces multicollinearity by selecting including or eliminating predictor variables with high correlation from the fitted model.

Regularized regression models were implemented using the `glmnet` package. LASSO (L1 regularization, $\alpha = 1$), ridge (L2 regularization, $\alpha = 0$), and ELNR ($\alpha = 0.5$) models were tuned over a grid of regularization parameters (`lambda` $= 10^{\{-3,0\}}$ with 10 values). The objective function for these models is:

$$\min_{\beta} \left[\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \left(\alpha \sum_{j=1}^{p} |\beta_j| + \frac{1-\alpha}{2} \sum_{j=1}^{p} \beta_j^2 \right) \right], \tag{4}$$

Where λ controls the strength of regularization, and α balances L1 and L2 penalties.

3.5.6. Hybrid ensemble model

A hybrid model was developed by combining predictions from LASSO, GBM, and RF models. Predictions from LASSO and GBM were added as features to the training and testing sets, and an RF model was trained on this augmented dataset. The hyperparameter `mtry` was tuned as in the standalone RF model.

Let:

- $X \in \mathbb{R}^{n \times p}$: original feature matrix
- $y \in \mathbb{R}^{n}$: observed crop yield
- $\hat{y}^{\text{LASSO}} = f_{\text{LASSO}}(X)$: predictions from LASSO regression
- $\hat{y}^{\text{GBM}} = f_{\text{GBM}}(X)$: predictions from Gradient Boosting
- $X^{\text{stack}} = [X \;\; \hat{y}^{\text{LASSO}} \;\; \hat{y}^{\text{GBM}}]$: stacked feature matrix
- $f_{\text{RF}}(\cdot)$: RF model trained on stacked features

Then, the final hybrid prediction is given by:

$$\hat{y}^{\text{Hybrid}} = f_{\text{RF}}(X^{\text{stack}}) = f_{\text{RF}}\left([X \;\; f_{\text{LASSO}}(X) \;\; f_{\text{GBM}}(X)]\right) \tag{5}$$

The random forest hyperparameter `mtry` was tuned using grid search over:

$$\texttt{mtry} \in \left\{ 2, \left\lfloor \frac{p+2}{3} \right\rfloor, p+1 \right\}$$

Where p is the number of original predictors in X, and the additional two features correspond to the LASSO and GBM predictions.

3.6. Performance evaluation

Model performance was evaluated using three metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2). These are defined as:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|,$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2},$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2},$$

Algorithm 1 Hybrid Ensemble Model (LASSO + GBM + RF on stacked predictions).

Require: Training data `train`, Testing data `test`
Require: Base models: LASSO, GBM; Meta-model: Random Forest (RF)
Require: Cross-validation control: `cv_control`
1: Train LASSO model on `train`
2: Train GBM model on `train`
3: Generate base predictions:
- `lasso_pred_train` ← LASSO predictions on `train`
- `gbm_pred_train` ← GBM predictions on `train`
- `lasso_pred_test` ← LASSO predictions on `test`
- `gbm_pred_test` ← GBM predictions on `test`

4: Augment datasets:
- `train_hybrid` ← `train` + `lasso_pred_train`, `gbm_pred_train`
- `test_hybrid` ← `test` + `lasso_pred_test`, `gbm_pred_test`

5: Define grid for RF hyperparameter `mtry`:

$$\texttt{mtry} \in \left\{ 2, \left\lfloor \frac{p+2}{3} \right\rfloor, p+1 \right\}$$

6: Train RF model on `train_hybrid` using 500 trees and `cv_control`
7: Predict yield on `test_hybrid` using trained RF model
8: Evaluate model performance using MAE, RMSE, R^2, and MAPE
9: **return** Final predictions and evaluation metrics

$$MAPE = \frac{100}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

Where $\bar{y}$ is the mean of the observed yields. These metrics were computed on the test set to assess model accuracy and goodness of fit.

3.7. District-level heterogeneity testing

To formally test district-level differences in rice yields, we applied a one-way analysis of variance (ANOVA) across 14 districts, followed by Tukey's Honest Significant Difference (HSD) post-hoc comparisons. To evaluate whether policy effects varied by district, a two-way ANOVA (District + Policy) was estimated. Finally, a mixed-effects model with random intercepts for districts was fitted to account for the panel structure of the data and unobserved spatial heterogeneity.

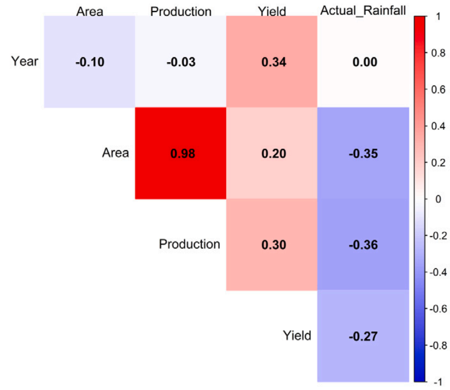
3.8. Implementation details

All analyzes were conducted in R (version 4.2.0 or later) using packages `tidyverse`, `caret`, `randomForest`, `gbm`, `e1071`, `glmnet`, `Metrics`, `psych`, `kernlab`, and `ggplot2`. A fixed seed (`set.seed(123)`) was used for model training to ensure reproducibility.

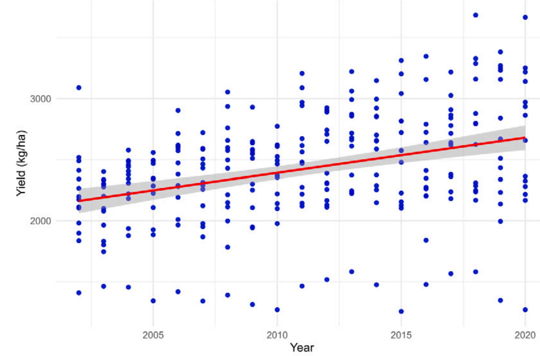
Correlation Analysis of Rice Yield Data:
To investigate the relationships among key variables influencing rice production, Pearson correlation coefficients were calculated for the numeric variables: year, area (cultivated land in hectares), production (total rice production in metric tons), yield (rice yield in kg/ha), and actual rainfall (annual rainfall in mm). The analysis was conducted using data from 266 observations spanning multiple districts and years (2002–2020). The Pearson correlation coefficient measures the strength and direction of linear relationships between pairs of variables, ranging from -1 to +1, with 0 indicating no linear relationship. Significance testing was performed to determine whether correlations were statistically significant at the 0.05 level, with p-values adjusted for multiple comparisons to control for type I errors.

A heatmap to display the Pearson correlation coefficients, with color intensity indicating the strength and direction of correlations. Scatter plots for key significant correlations for area vs. production, yield vs. actual rainfall, year vs. yield to provide a detailed view of the relationships.

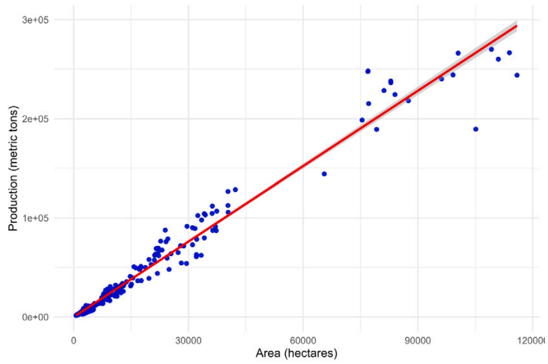
Fig. 4a the heatmap illustrates the Pearson correlation coefficients among year, area, production, yield, and actual rainfall. Red shades indicate positive correlations, blue shades indicate negative correlations, and color intensity reflects the strength of the correlation. Fig. 4b



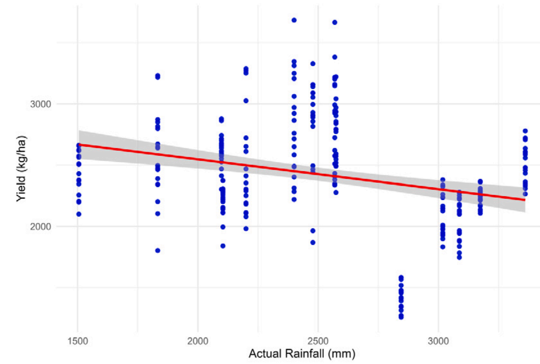
(a) Correlation matrix heatmap of Year, Area, Production, Yield, and Actual Rainfall



(b) Scatter plot of Year vs. Yield, showing a positive correlation (r = 0.34, p < 0.01).



(c) Scatter plot of Area vs. Production, showing a strong positive correlation (r = 0.98, p < 0.01)



(d) Scatter plot of Actual Rainfall vs. Yield, showing a negative correlation (r = -0.27, p < 0.01).

Fig. 4. Correlation Analysis.

the significant positive correlation between year and yield suggests improvements in rice yield over time, potentially due to advancements in agricultural technology or practices. This finding highlights the importance of continued investment in research and development to sustain and enhance yield improvements.

Fig. 4c shows very strong and highly significant positive correlation was observed between area and production ($r = 0.98$, $p < 0.01$). This near-perfect linear relationship indicates that increases in cultivated area are strongly associated with increases in rice production. This finding is expected, as larger cultivated areas typically allow for greater total production, assuming consistent yield rates. The strength of this correlation underscores the importance of land availability as a primary driver of rice production in the region. Fig. 4d actual rainfall exhibited significant negative correlations yield ($r = -0.27$, $p < 0.01$). The negative correlations suggest that higher rainfall levels are associated with reduced cultivated area, lower total production, and decreased yields. This finding may reflect the adverse effects of excessive rainfall, such as flooding or waterlogging, which can damage rice crops and reduce productivity. The moderate strength of these correlations indicates that while rainfall is an important factor, other environmental or management factors also influence these outcomes.

RF Variable Importance:

To understand the relative influence of each predictor variable on rice yield, a variable importance analysis was conducted using the Random Forest model. The importance was measured using the Mean Decrease in Gini index, which quantifies the contribution of each variable to the homogeneity of the nodes and leaves in the decision trees.

The Fig. 5 reveal that actual rainfall was the most influential variable, with the highest importance score of 9,259,302.1, underscoring the critical role of climatic factors in determining rice productivity in the region. This aligns with established agronomic knowledge that rainfall variability directly affects water availability during key phenological stages of rice, thereby impacting yield.

The next most important variables were district Kozhikode (8,217, 762.8), production (5,155,660.6), year (5,053,227.9), and area (3,765, 737.5). The high importance of district Kozhikode suggests a strong spatial heterogeneity in yield patterns, potentially attributable to regional agro-climatic or soil characteristics. Similarly, the prominence of production and area reflects their expected direct association with yield, especially when computed as yield = production/area. Interestingly, year also emerged as a strong predictor, indicating that inter-annual variability possibly due to changing weather patterns, policy interventions, or technology adoption plays a substantial role in yield dynamics. In contrast, several district-level dummy variables district Palakkad, Wayanad exhibited relatively lower importance scores, suggesting that while location is a relevant factor, its predictive power varies significantly across regions.

We built a linear regression model using yield (kg/ha) as the dependent variable and actual rainfall and a binary policy indicator as independent variables. The policy variable is 1 for years 2010 and after and 0 for years before that.

$$\text{Yield}_{it} = \beta_0 + \beta_1 \cdot \text{Rainfall}_{it} + \beta_2 \cdot \text{Policy}_{it} + \epsilon_{it}$$

where i denotes the district and t denotes the year.

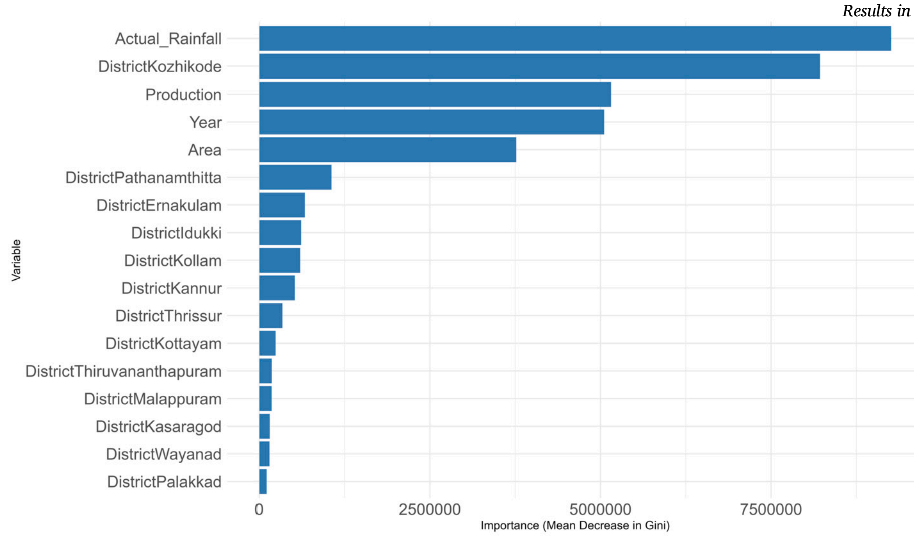


Fig. 5. Variable Significance in Random Forest: Insights into Model Decision-Making.

Table 4
Performance Metrics of Machine Learning Models for Rice Yield Prediction.

Model	R^2	MAE	RMSE	MAPE (%)
Random Forest	0.841	128.808	181.173	5.15
Gradient Boosting	0.893	111.233	148.784	4.46
Support Vector Regression	0.857	118.586	169.810	4.77
Multiple Linear Regression	0.852	135.244	173.665	5.56
Lasso Regression	0.849	135.493	175.126	5.58
Ridge Regression	0.793	155.078	205.715	6.34
Elastic Net Regression	0.850	135.527	174.562	5.58
Hybrid Model	**0.918**	**92.518**	**130.506**	**3.67**

4. Results & discussion

To evaluate the performance of the machine learning models for predicting rice yield, we employed eight models: RF, GBR, SVR, MLR, LASSO, RR, ELNR, and a hybrid model combining RF and GB. The models were assessed using three metrics: R^2, MAE, RMSE and MAPE. These metrics provide insights into the models' explanatory power and predictive accuracy. The performance metrics are summarized in Table 4.

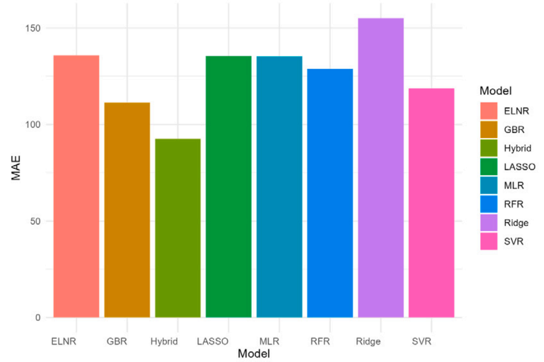
The hybrid model achieved the highest performance across all metrics, with an R^2 of 0.918, MAE of 92.518 kg/ha, and RMSE of 130.506 kg/ha. This indicates that the hybrid model explains approximately 91.8% of the variance in rice yield and provides the most accurate predictions, with the lowest average error. GBR followed closely, with an R^2 of 0.893, MAE of 111.233 kg/ha, and RMSE of 148.784 kg/ha, demonstrating strong predictive capability. SVR and MLR also performed well, with R^2 values of 0.857 and 0.852, respectively, and comparable MAE and RMSE values. LASSO and ELNR showed similar performance, with R^2 values around 0.85 and slightly higher errors than SVR and MLR. RF performed adequately (R^2 = 0.841), but its errors were higher than those of GB, SVR, and MLR. Ridge regression exhibited the lowest performance, with an R^2 of 0.793, MAE of 155.078 kg/ha, and RMSE of 205.715 kg/ha, indicating reduced explanatory power and higher prediction errors (Fig. 6).

To ensure accuracy and practicality, the scatter plot is a crucial tool for testing predictive models. This visual technique, along with statistical measures, creates a strong method for examining and improving model effectiveness across various applications. The scatter plot illustrated compares anticipated values to actual data points, offering a straightforward visualization of model performance. Each point on the plot signifies a specific prediction made by the model compared to its relevant actual observed value. This facilitates an immediate visual evaluation of how well the model's predictions align with reality. In Fig. 7, the distribution and density of points along this diagonal reflect the model's precision and potential biases. A model that has a higher R^2 value and data points that are nearer to the diagonal is regarded as the best. Models with a lower R^2 also show a higher degree of dispersion. A close grouping of points along the diagonal suggests high precision, indicating that the model's predictions closely resemble the observed data. Deviations from this line reveal areas where the model may underpredict or overpredict.

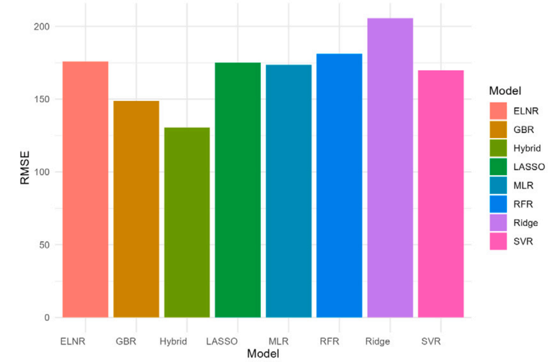
The results demonstrate that the hybrid model, which combines the strengths of RF and GBR, outperforms other models in predicting rice yield, achieving an R^2 of 0.918 and the lowest errors (MAE = 92.518 kg/ha, RMSE = 130.506 kg/ha). The superior performance of the hybrid model can be attributed to its ability to leverage the robustness of RF in handling non-linear relationships and the iterative optimization of gradient boosting in minimizing prediction errors. This finding aligns with previous studies, such as those by [6] and [10], which highlight the efficacy of ensemble methods in complex regression tasks. The hybrid model's low MAE and RMSE suggest it is particularly suitable for practical applications, such as precision agriculture, where accurate yield predictions can inform resource allocation and policy decisions.

Gradient boosting also performed exceptionally well (R^2 = 0.893), outperforming SVR and linear models. This is consistent with the findings of [8], who noted that boosting methods excel in capturing complex interactions among predictors, such as those between area, production, and actual rainfall observed in our correlation analysis. The strong correlation between area and production ($r = 0.98$, $p < 0.01$) likely contributed to the models' ability to predict yield accurately, as area is a primary driver of total production, which is closely related to yield ($r = 0.30$, $p < 0.01$). However, the negative correlation between actual rainfall and yield ($r = -0.27$, $p < 0.01$) suggests that excessive rainfall may reduce yields, possibly due to flooding or waterlogging, which the non-linear models (Hybrid, GB, RF, SVR) were better equipped to capture compared to linear models.

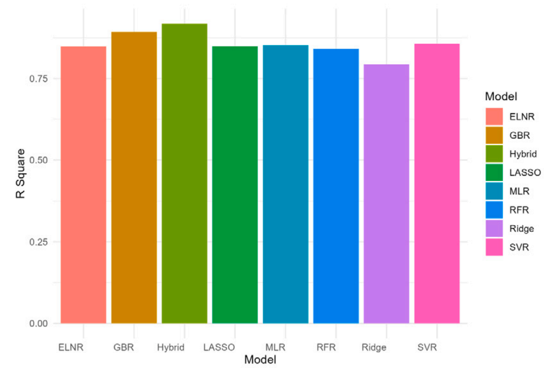
SVR (R^2 = 0.857) and MLR (R^2 = 0.852) showed comparable performance, indicating that both linear and non-linear approaches can effectively model rice yield in this dataset. SVR's ability to handle non-linear relationships through kernel functions likely contributed to its slight edge over MLR in terms of RMSE (169.810 kg/ha vs. 173.665 kg/ha). LASSO and ELNR, which incorporate regularization to prevent overfitting, performed similarly to MLR but did not surpass SVR or GB, possibly due to the dataset's moderate size ($n = 266$) and the presence of non-linear relationships that regularization alone could not fully address. Ridge regression's lower performance (R^2 = 0.793) may be attributed to its sensitivity to multicollinearity, particularly between



(a) Mean Absolute Error

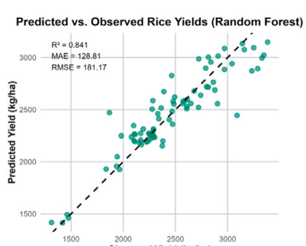


(b) Root Mean Square Error

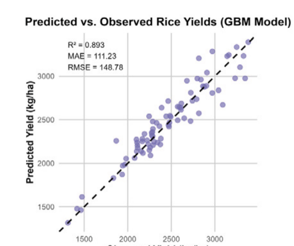


(c) R-squared Value

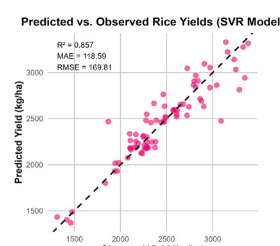
Fig. 6. Performance metrics across regression models.



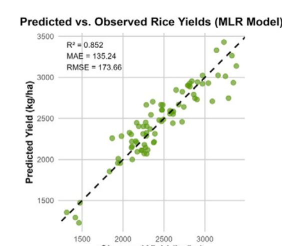
(a) Random Forest



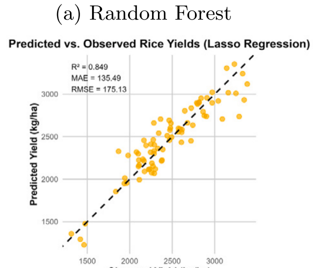
(b) Gradient Boosting



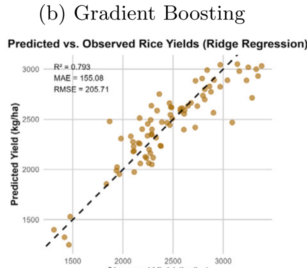
(c) SV Regression



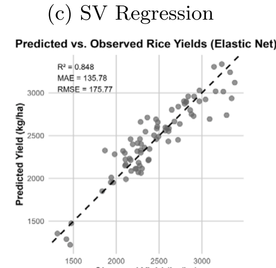
(d) Linear Model



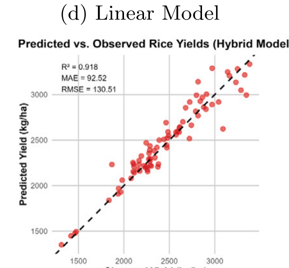
(e) LASSO Model



(f) Ridge Model



(g) Elastic Net



(h) Hybrid Model

Fig. 7. Comparison of predicted and actual crop yield across regression models.

area and production, which regularized linear models like LASSO and ELNR better mitigated.

Although the hybrid ensemble achieved the highest predictive accuracy (R^2 = 0.918), the gain relative to GBR (R^2 = 0.893) appears modest in absolute terms (<3%). However, the ensemble consistently outperformed individual models across multiple evaluation metrics, including reductions in RMSE (148.8 → 130.5 kg/ha), MAE (111.2 → 92.5 kg/ha), and MAPE (4.5% → 3.7%). When contrasted with SVR (R^2 = 0.837) and MLR (R^2 = 0.812), the ensemble's superiority is more high, underscoring its practical utility. These results highlight that the ensemble's value is not only in marginal R^2 improvement but in providing stable, reliable predictions critical for policy applications.

To assess the generalizability of the hybrid ensemble model, a learning curve was generated by evaluating performance across incremental training set sizes Fig. 8. Both RMSE and MAE showed consistent declines as the training size increased, indicating improved predictive

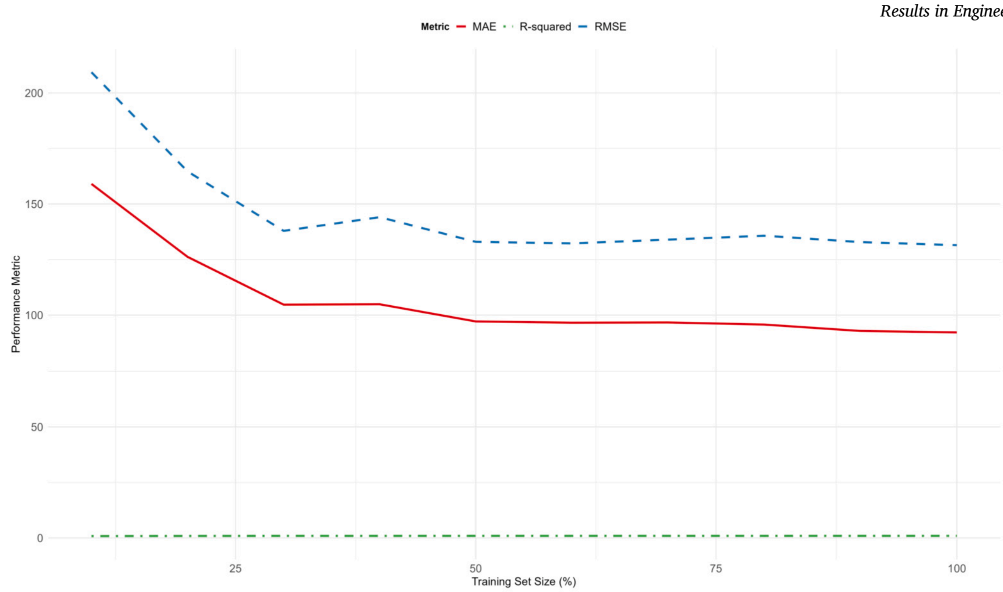


Fig. 8. Learning curve showing RMSE, MAE, and R-squared performance of the hybrid model across increasing training set sizes.

accuracy and reduced variance. In contrast, the R-squared metric remained relatively stable, suggesting that the model's explanatory power was saturated early on. These trends confirm that the hybrid model is well-regularized and not overfitting, despite the modest sample size ($n = 266$). The learning curve also highlights the model's sensitivity to data volume, particularly in the lower range of training sizes, where performance gains were most pronounced.

The moderate positive correlation between year and yield (r = 0.34, p < 0.01) suggests a temporal increase in yield, likely due to advancements in agricultural practices, such as improved seed varieties or irrigation techniques. This trend was effectively captured by the hybrid and GBR models, which likely weighted temporal features appropriately. The negative correlation between actual rainfall and yield highlights the importance of water management in rice cultivation. Excessive rainfall, as observed in the dataset (mean = 2517.57 mm, SD = 521.09 mm), may lead to adverse effects, such as reduced yields due to water stress. This finding is consistent with [26], who reported that excessive rainfall can negatively impact rice productivity in tropical regions.

The regression analysis revealed that policy intervention was significantly associated with increased crop yield. The estimated coefficient for the policy variable was +279.15 kg/ha (SE = 52.44, $p < 0.001$), indicating a substantial positive effect in the post-intervention period. This relationship held after controlling for climatic factors, specifically actual rainfall. Rainfall exhibited a significant negative association with yield, with an estimated effect of −0.243 kg/ha per mm (SE = 0.050, $p < 0.001$). This suggests that excessive rainfall may have contributed to reduced yields, potentially due to agronomic stresses such as waterlogging or nutrient leaching.

The findings suggest that rainfall is the most influential factor shaping rice yields in Kerala, underscoring the vulnerability of rainfed districts such as Wayanad and Palakkad. Accordingly, policy interventions should prioritize these regions by expanding irrigation infrastructure, promoting micro-irrigation practices, and encouraging the adoption of climate-resilient rice varieties. Moreover, policies should adopt a district-specific approach, targeting resource allocation and technology dissemination to areas where yield variability is most strongly driven by climatic shocks. Such targeted strategies would enhance resilience, rather than relying on uniform state-level policies.

4.1. Spatial autocorrelation and district-level heterogeneity

Rainfall emerged as the most influential predictor in the hybrid ensemble model, consistent with Kerala's monsoon-dependent rice ecosystems. However, this dominance was not spatially uniform. Districts such as Kozhikode and Wayanad exhibited higher relative importance scores compared to southern districts, indicating spatial heterogeneity in the rainfall–yield relationship.

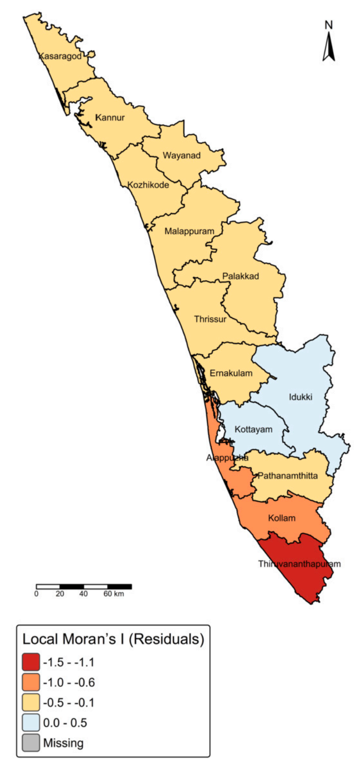


Fig. 9. Local Moran's I cluster map of residuals from the hybrid ensemble model, showing absence of significant spatial clustering across districts.

To assess whether residual spatial dependencies were present, we conducted a Global Moran's I test on the model residuals. The resulting Moran's I statistic was –0.238 (p = 0.797), suggesting that residuals were spatially uncorrelated. This implies that the model adequately captured spatial structure and that rainfall's predictive dominance was not confounded by unobserved district-level clustering.

Table 5
District-level heterogeneity in rice yields: ANOVA and mixed-effects results.

Model/Test	Statistic	df	p-value
One-way ANOVA (District)	$F = 39.45$	(13, 252)	< 0.001
Two-way ANOVA (District + Policy)	District: $F = 54.04$	(13, 251)	< 0.001
	Policy: $F = 94.21$	(1, 251)	< 0.001
Mixed-effects model	Variance (District) = 143,811	–	–
	Policy effect (β = 279.1, SE = 28.8, $t = 9.71$)	–	< 0.001
	Rainfall effect (β = -0.24, SE = 0.20, $t = -1.23$)	–	0.22

Additionally, a Local Moran's I cluster map Fig. 9 was generated to visualize district-level residual patterns. No significant spatial clusters were detected, further supporting the robustness of the variable importance interpretation.

Table 5 summarizes the ANOVA and mixed-effects findings. A one-way ANOVA revealed significant heterogeneity in rice yields across districts (F (13,252) = 39.45, p < 0.001). Post-hoc Tukey tests identified Kozhikode, Kannur, and Kasaragod as consistently lower-yielding compared to districts such as Alappuzha, Thrissur, and Pathanamthitta (mean differences exceeding 600–1,300 kg/ha, p < 0.001). In contrast, central districts such as Alappuzha, Kottayam, and Pathanamthitta did not differ significantly (p < 0.9), suggesting regional clustering of productivity.

In a two-way ANOVA, both district (F (13,251) = 54.04, p < 0.001) and policy (F (1,251) = 94.21, p < 0.001) effects were statistically significant. This indicates that while the 2010 policy intervention improved yields overall, the magnitude of gains differed by district.

The mixed-effects model, incorporating random district intercepts, confirmed significant between-district variability (variance = 143,811; SD = 379.2). The policy effect remained positive and robust (β = 279.1, SE = 28.8, t = 9.71), while rainfall showed weaker, non-significant effects (β = –0.24, SE = 0.20, t = –1.23). These results highlight that baseline productivity levels vary considerably across districts, but the policy intervention had a consistent positive effect across the state.

A line plot demonstrated a clear upward trend in mean yields following the policy's implementation in 2010 Fig. 10a. This plot illustrated a negative linear relationship between rainfall and yield, with notable heterogeneity between pre- and post-policy periods Fig. 10b. District-wise comparisons revealed non-uniform yield gains, with the largest improvements in traditionally rainfed districts such as Wayanad and Palakkad Fig. 10c. The Fig. 10 below indicated the statistical models and gave us a deeper understanding of how regional yields change over time.

Our results highlight pronounced spatial heterogeneity in rice yields across Kerala. District-level ANOVA and mixed-effects modeling confirmed that agro-ecological and structural factors create significant productivity gaps, with some districts consistently outperforming others. Importantly, the 2010 policy intervention was associated with yield improvements across all districts, although the magnitude of the effect varied. This underscores the need for spatially differentiated policy strategies, tailored to the constraints and opportunities of high- and low-yielding regions.

It is important to note that the associations observed between policy variables and yield outcomes should not be interpreted as definitive causal effects. While our models demonstrate strong predictive capacity and identify policies as statistically significant correlates of yield growth, the absence of finer control variables (seed varieties, fertilizer intensity, pest management) limits strict causal inference. Thus, our findings are best understood as evidence of robust associations rather than proof of direct policy causation.

4.2. Limitations and future directions

A key limitation of this study is the reliance on district-level aggregated data, which does not capture intra-district variability such as soil fertility gradients, pest outbreaks, or micro-climatic fluctuations. While recent studies [4] have employed climate–NDVI fusion to integrate finer-resolution data, such datasets were not uniformly available for the full study period (2002–2020).

Future research should incorporate additional biophysical and socio-economic variables such as soil fertility, pest incidence, and access to agricultural inputs, and should also evaluate model performance across diverse agro-climatic regions. Moreover, extending the current framework to higher-resolution spatio-temporal data would enable more granular insights into yield dynamics and regional heterogeneity.

Methodologically, future studies could benefit from adopting advanced causal frameworks such as Difference-in-Differences (DID), fixed-effects panel models, or synthetic control approaches to better isolate policy impacts while accounting for time-varying confounders. Furthermore, the integration of Functional Data Analysis (FDA) could capture temporal evolution in climatic and agronomic variables, allowing for dynamic yield modeling and improved understanding of crop response trajectories [22,15].

Overall, this study contributes to the growing body of literature by proposing a scalable, policy-relevant analytical framework for environmentally sustainable crop yield forecasting and data-driven food system planning.

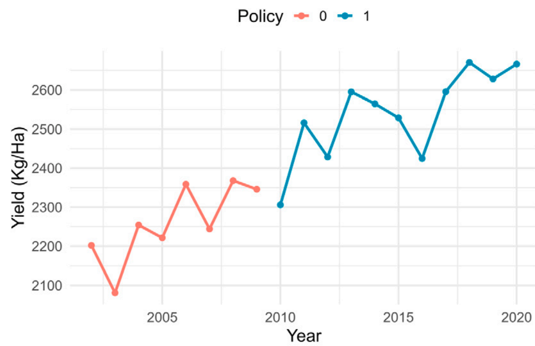
5. Conclusion & future scope

This study demonstrates that machine learning approaches can significantly enhance the accuracy of rice yield predictions in Kerala, India. The proposed hybrid ensemble model yielded the most reliable results ($R^2 = 0.918$, $MAE = 92.518\ kg/ha$, $RMSE = 130.506\ kg/ha$), outperforming other regression-based models. A strong statistical relationship was observed between the cultivated area and crop output ($r = 0.98$, $p < 0.01$), while rainfall exhibited a negative correlation with yield ($r = -0.27$, $p < 0.01$), emphasizing the climatic sensitivity of rice production.

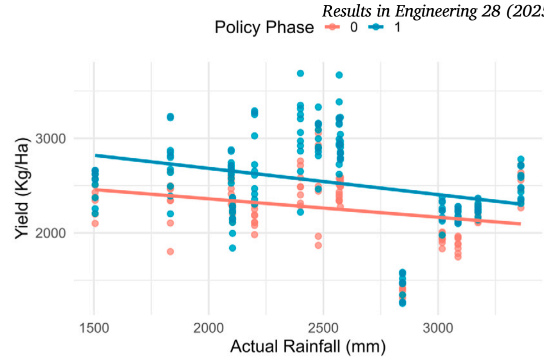
A regression-based policy model further revealed that post-2010 policy interventions were associated with a statistically significant improvement in yield, underscoring the potential benefits of integrating policy insights with predictive analytics. These findings highlight the dual importance of precise modeling and institutional support in promoting agricultural resilience and sustainability.

Our results align with and complement prior research. For instance, Satpathi et al. [20] reported that ANN can outperform other machine learning models for yield forecasting, though our study avoided ANNs to mitigate overfitting risks associated with limited district-level data. Instead, we prioritized the interpretability and robustness of regression-based ensembles such as Random Forest, Gradient Boosting, and the proposed Hybrid model. Similarly, Guo et al. [13] emphasized the predictive importance of phenological variables (e.g., sowing and flowering dates), which were unavailable in our dataset, thereby constraining the current analysis to rainfall, irrigation, and production indicators.

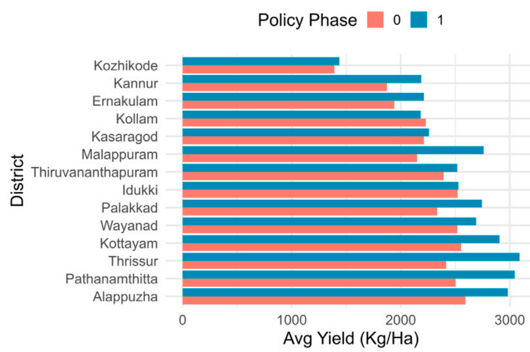
Within the Indian context, Yumlembam et al. [31] employed linear regression for rice yield forecasting in Manipur and identified sunshine hours as the most influential factor, linking reduced rainfall to potential threats to cultivation. In contrast, our findings position rainfall as the dominant driver, reflecting regional agro-ecological variability in yield determinants. Taken together, these comparisons highlight how model selection, data resolution, and variable availability explain differences



(a) Average Yield Over Time



(b) Pre and Post-Policy



(c) District-Level Yield Comparison

Fig. 10. Impact of Policy Intervention on Crop Yield Dynamics.

in predictive outcomes underscoring the complementarity rather than contradiction of our findings with prior research.

CRediT authorship contribution statement

Saruk B S: Writing – original draft, Visualization, Software, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Mokesh Rayalu G:** Writing – review & editing, Validation, Conceptualization.

Ethics approval and consent to participate

This manuscript does not contain any studies with human participants or animals performed by the author.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Upon a reasonable request, the sources and other pertinent information will be provided.

References

[1] Md Sakib, Bin Alam, Vatcharaporn Esichaikul, Aiman Lameesa, Shams Forruque Ahmed, Amir H. Gandomi, An approach for crop recommendation with uncertainty quantification based on machine learning for sustainable agricultural decision-making, Results Eng. (2025) 105505.
[2] A.M. Ali, H.S. Thind, S. Sharma, et al., Prediction of dry direct-seeded rice yields using chlorophyll meter, leaf color chart and greenseeker optical sensor in northwestern India, Field Crops Res. 161 (2014) 11–15.
[3] Wisdom Aselisewine, Suvra Pal, Enhancing cure rate analysis through integration of machine learning models: a comparative study, Stat. Comput. 34 (4) (2024) 142.
[4] Muhammad Ashfaq, Imran Khan, Abdulrahman Alzahrani, Muhammad Usman Tariq, Humera Khan, Anwar Ghani, Accurate wheat yield prediction using machine learning and climate-ndvi data fusion, IEEE Access 12 (2024) 40947–40961.
[5] J.W. Bauder, G.W. Randall, Regression models for predicting corn yields from climatic data and management practices, Soil Sci. Soc. Am. J. 46 (1) (1982) 158–161.
[6] Leo Breiman, Random forests, Mach. Learn. 45 (2001) 5–32.
[7] Girish Chandrashekar, Ferat Sahin, A survey on feature selection methods, Comput. Electr. Eng. 40 (1) (2014) 16–28.
[8] Tianqi Chen, Carlos Guestrin, Xgboost: a scalable tree boosting system, in: Proceedings of the 22nd Acm Sigkdd International Conference on Knowledge Discovery and Data Mining, 2016, pp. 785–794.
[9] Andrew Crane-Droesch, Machine learning methods for crop yield prediction and climate change impact assessment in agriculture, Environ. Res. Lett. 13 (11) (2018) 114003.
[10] Jerome H. Friedman, Greedy function approximation: a gradient boosting machine, Ann. Stat. (2001) 1189–1232.
[11] Niketa Gandhi, Leisa J. Armstrong, Owaiz Petkar, Amiya Kumar Tripathy, Rice crop yield prediction in India using support vector machines, in: 2016 13th International Joint Conference on Computer Science and Software Engineering (JCSSE), IEEE, 2016, pp. 1–5.
[12] Antoine Godichon-Baggioni, Wei Lu, Bruno Portier, Recursive ridge regression using second-order stochastic algorithms, Comput. Stat. Data Anal. 190 (2024) 107854.

[13] Yahui Guo, Yongshuo Fu, Fanghua Hao, Xuan Zhang, Wenxiang Wu, Xiuliang Jin, Christopher Robin Bryant, J. Senthilnath, Integrated phenology and climate in rice yields prediction using machine learning methods, Ecol. Indic. 120 (2021) 106935.
[14] MD Jiabul Hoque, Md Saiful Islam, Jia Uddin, Md Abdus Samad, Beatriz Sainz De Abajo, Débora Libertad Ramírez Vargas, Imran Ashraf, Incorporating meteorological data and pesticide information to forecast crop yields using machine learning, IEEE Access (2024).
[15] Hidetoshi Matsui, Keiichi Mochida, Functional data analysis-based yield modeling in year-round crop cultivation, Hortic. Res. 11 (7) (2024) uhae144.
[16] Meera G. Mohan, S. Adarsh, Statistical characterization of maximum temperature of kerala, India, in: AIJR Proceedings, 2023, pp. 120–128.
[17] Bharati Panigrahi, Krishna Chaitanya Rao Kathala, M. Sujatha, A machine learning-based comparative approach to predict the crop yield using supervised learning with regression models, Proc. Comput. Sci. 218 (2023) 2684–2693.
[18] Robi Polikar, Ensemble learning, in: Ensemble Machine Learning: Methods and Applications, 2012, pp. 1–34.
[19] B.S. Saruk, G. Mokesh Rayalu, Integrating machine learning for enhanced agricultural productivity: a focus on bananas and arecanut in the context of India's economic growth, J. Stat. Theory Appl. (2024) 1–19.
[20] Anurag Satpathi, Parul Setiya, Bappa Das, Ajeet Singh Nain, Prakash Kumar Jha, Surendra Singh, Shikha Singh, Comparative analysis of statistical and machine learning techniques for rice yield forecasting for Chhattisgarh, India, Sustainability 15 (3) (2023) 2786.
[21] Parul Setiya, Ajeet Singh Nain, Anurag Satpathi, Comparative analysis of smlr, ann, elastic net and lasso based models for rice crop yield prediction in Uttarakhand, Mausam 75 (1) (2024) 191–196.
[22] Ismail Shah, Naveed Gul, Sajid Ali, Hassan Houmani, Short-term hourly ozone concentration forecasting using functional data approach, Econometrics 12 (2) (2024) 12.
[23] Aditya Shastry, H.A. Sanjay, E. Bhanusree, Prediction of crop yield using regression techniques, Int. J. Soft Comput. 12 (2) (2017) 96–102.
[24] Wenjiao Shi, Fulu Tao, Zhao Zhang, A review on statistical models for identifying climate contributions to crop yields, J. Geogr. Sci. 23 (2013) 567–576.
[25] U. Surendran, P. Raja, M. Jayakumar, S. Rama Subramoniam, Use of efficient water saving techniques for production of rice in India under climate change scenario: a critical review, J. Clean. Prod. 309 (2021) 127272.
[26] Reiner Wassmann, S.V.K. Jagadish, K. Sumfleth, H. Pathak, G. Howell, A. Ismail, Rachid Serraj, E. Redona, R.K. Singh, S. Heuer, Regional vulnerability of climate change impacts on Asian rice production and scope for adaptation, Adv. Agron. 102 (2009) 91–133.
[27] Lasini Wickramasinghe, Rukmal Weliwatta, Piyal Ekanayake, Jeevani Jayasinghe, Modeling the relationship between rice yield and climate variables using statistical and machine learning techniques, J. Math. 2021 (1) (2021) 6646126.
[28] Shizhong Xu, Yang Xu, Liang Gong, Qifa Zhang, Metabolomic prediction of yield in hybrid rice, Plant J. 88 (2) (2016) 219–227.
[29] Li Yang, Abdallah Shami, On hyperparameter optimization of machine learning algorithms: theory and practice, Neurocomputing 415 (2020) 295–316.
[30] Stephen Luo Sheng Yong, Jing Lin Ng, Yuk Feng Huang, Chun Kit Ang, Norashikin Ahmad Kamal, Majid Mirzaei, Ali Najah Ahmed, Enhanced daily reference evapotranspiration estimation using optimized hybrid support vector regression models, Water Resour. Manag. 38 (11) (2024) 4213–4241.
[31] Janee Yumlembam, Yumlembam Chakrabarty Singh, A.K. Nandi, Effect of weather parameters on rice productivity in manipur, Indian J. Econ. Dev. 13 (4) (2017) 735–740.